SOCIOL 723 Social Statistics II
Estimation, Causal Inference, and Machine Learning

Time and Location

Fall 2026
Tuesdays (Lecture) 10:05AM–12:35PM; Thursdays (Lab) 9:05–11:35AM
Erwin Square Plaza Room 753

Instructor Information

Professor Wenhao Jiang
Email wenhao.jiang@duke.edu
Office Hours Wednesdays 2–4PM or by appointment

Course Overview and Prerequisites

This is the second course in the graduate statistics sequence for sociology doctoral students. *Social Statistics I*, using the model-comparison approach of Correll et al., rigorously developed regression, ANOVA, and a first look at logistic and Poisson regression in `R`. Building on that foundation, this course extends into more advanced quantitative methodology, organized around three pillars: estimation, causal inference, and machine learning. We begin by revisiting ordinary least squares (OLS), recasting the model-comparison framework in matrix form and attending carefully to its statistical properties, its interpretation as a tool for causal adjustment, and inference under realistic assumptions, including heteroskedasticity- and cluster-robust standard errors. We then develop maximum likelihood estimation (MLE) as the general engine behind most models sociologists use, grounding the logistic and Poisson models introduced in *Social Statistics I* in likelihood theory and extending them, within the generalized linear model framework, to multinomial, ordered, and other limited dependent variables.

The core of the course turns to causal inference, the central methodological concern of contemporary empirical social science. We introduce the potential outcomes framework and directed acyclic graphs (DAGs) as complementary languages for defining and identifying causal effects, then work through the standard research designs: selection on observables (matching, propensity scores, and weighting), instrumental variables (IV), regression discontinuity designs (RDD), difference-in-differences (DID) with panel data, and synthetic control. The course closes by introducing machine learning for prediction and its integration with causal inference, including double/debiased machine learning and the estimation of heterogeneous treatment effects.

Each week pairs a Tuesday lecture that develops the theory with a Thursday lab that implements the methods on real data in `R`. By the end of the course, students should be

able to read and critically evaluate the quantitative sociological literature that uses these methods, and to apply them competently in their own research. The course assumes working knowledge of the general linear model and statistical inference at the level of *Social Statistics I*. Familiarity with matrix algebra and calculus is helpful but not required; everything needed is introduced from zero in the first two weeks. All coursework is in `R`.

Expectations

Reading and participation (10%):
You are expected to complete the required readings before the class in which they are discussed and to participate actively in both lecture and lab. If a reading is difficult, come prepared to discuss what you did and did not follow.

Problem sets (30%):
There will be 5 problem sets, assigned roughly bi-weekly. They reinforce the lecture material and typically require `R` coding building on that week's lab. Problem sets are generally due two weeks after release, by 11:59PM (see the schedule for exact dates).

In-Class Midterm Exam (25%):
There will be an in-class midterm on **Thursday, October 15**, covering the estimation half of the course (OLS and MLE) and the foundations of causal inference through instrumental variables (Weeks 1–7). The exam is open book and open note but timed. You will not be required to write any code.

Final Assessment (35%):
There will be a cumulative final exam on **Thursday, December 3, 9:05–11:35AM**. It covers the whole course, but weighted towards **Weeks 9–13**, since the midterm already examined Weeks 1–7. Like the midterm, it is open book and open note but timed, and you will not be required to write any code. What is examined is the material developed in scalar form, together with what each design assumes and what it estimates; frames marked with a blue star in the slides flag the essential material the exams draw on, while frames marked with an orange star are for understanding and will not be examined.

You may complete the final assessment in either of two ways:

- **Option A, exam only**: the final exam counts for the full 35%.
- **Option B, exam plus project**: the final exam counts for 25%, and a **final project memo** counts for the remaining 10%. The memo is 6–10 pages applying one of the methods developed in this course to your own research data (publicly available or not, either is fine). Submit the memo together with full replication code by **Friday, December 11, 11:59PM**.

Software

This class uses `R` throughout, and prior experience at the level of *Social Statistics I* is assumed. Please install a recent version of `R` and `RStudio` before the first lab, and use

`RMarkdown` or `Quarto` for your assignments to support reproducibility.

Use of Artificial Intelligence (AI)

I encourage you to use AI tools to support your statistical learning, particularly for working through steps in a derivation and for debugging `R` code. AI assistance is permitted on problem sets, provided you understand every step of what you submit and write it up in your own words.

AI is **not** permitted during either exam. That is deliberate, and it is why the exams carry the weight they do: they are the only points in the course where what is being assessed is unambiguously *your own* understanding.

Course Materials

We will draw regularly on three texts, two of which are free online.

- [Effect] Huntington-Klein, Nick. 2022. *The Effect: An Introduction to Research Design and Causality*. CRC Press. (online)
- [MHE] Angrist, Joshua D. and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press. (online)
- [CCI] Morgan, Stephen L. and Christopher Winship. 2015. *Counterfactuals and Causal Inference: Methods and Principles for Social Research*, 2nd ed. Cambridge University Press.

The Effect and the Chernozhukov et al. text are living online books whose chapter numbers shift between editions; chapter titles are given where the number alone might mislead. Within each week, the readings are listed *in the order you should read them*: the first builds the intuition, and those that follow supply the rigour. *Further reading* is there for depth and for your own research projects.

The remaining texts are used only for particular weeks and are freely available or optional rather than required purchases:

- Ward, Michael D. and John S. Ahlquist. 2018. *Maximum Likelihood for Social Science: Strategies for Analysis*. Cambridge University Press. (maximum likelihood, written for social scientists)
- [ISL] James, Gareth et al. 2021. *An Introduction to Statistical Learning: with Applications in R*, 2nd ed. Springer. (online)
- [Pawitan] Pawitan, Yudi. 2013. *In All Likelihood: Statistical Modelling and Inference Using Likelihood*. Oxford University Press.
- [Mixtape] Cunningham, Scott. 2021. *Causal Inference: The Mixtape*. Yale University Press. (online)
- [CML] Chernozhukov, Victor et al. 2025. *Applied Causal Inference Powered by ML and AI*. (online)

For review of prerequisite material, students may consult the text used in *Social Statistics I*: Correll, Josh, Abigail M. Folberg, Charles M. Judd, Gary H. McClelland, and Carey S. Ryan, *Data Analysis: A Model Comparison Approach to Regression, ANOVA, and Beyond* (Routledge), together with its `R` companion, Stephen Vaisey's *DAMCA*.

Schedule

The schedule is tentative as of November 6, 2026. A detailed list of topics and readings, by week, appears below. Tuesday sessions are lectures and Thursday sessions are `R` labs. You are responsible for completing the *required readings* before the Tuesday lecture.

Week	Dates	Topic	Problem sets	
			Assign	Due
1	Aug 25/27	The Linear Model and OLS		
2	Sep 1/3	Regression Inference and Robust Standard Errors	1	
3	Sep 8/10	Maximum Likelihood: Theory		
4	Sep 15/17	Maximum Likelihood: Applications; Multilevel Models		1
5	Sep 22/24	Causal Inference: Potential Outcomes and DAGs	2	
6	Sep 29/Oct 1	Matching, Propensity Scores, and Weighting		
7	Oct 6/8	Instrumental Variables		2
8	Oct 13/15	*Fall break* (Tues) and *In-class Midterm* (Thurs)		
9	Oct 20/22	Regression Discontinuity Designs	3	
10	Oct 27/29	Panel Data and Difference-in-Differences		
11	Nov 3/5	Synthetic Control and Its Extensions	4	3
12	Nov 10/12	Machine Learning: Prediction and Regularization		
13	Nov 17/19	Causal Machine Learning	5	4
14	Nov 24	Synthesis and Exam Review		
	Dec 3	*Cumulative final exam* (Thursday, 9:05–11:35AM)		5

Tuesday sessions are lectures; Thursday sessions are hands-on `R` labs applying the week's methods. Fall break is Oct 9–14 (no Tuesday class Oct 13); the Thanksgiving recess begins Nov 24, so there is no Thursday lab on Nov 26. Problem Set 4 (Weeks 10–11) is assigned Friday, Nov 6 and due Friday, Nov 20; Problem Set 5 (Weeks 12–13) is assigned Friday, Nov 20 and due Friday, Dec 4.

Week 1 (Aug 25/27): The Linear Model and OLS

Lecture: The general linear model in scalar and matrix form, connecting the model-comparison framework of *Social Statistics I* to matrix notation; the OLS estimator and its geometry; the Frisch–Waugh–Lovell theorem, omitted-variable bias, and regression as a tool for adjustment. *Lab*: `R` and `RStudio` refresher; OLS computed by hand in scalar form, on simulated data and the GSS.

- *Reading*: Effect, §13.1–13.2; MHE, §3.1–3.2.
- *Further reading*: CCI, Chapter 6; ISL, Chapter 3.

Week 2 (Sep 1/3): Regression Inference and Robust Standard Errors

Lecture: Sampling distribution of OLS and hypothesis testing; heteroskedasticity; robust and cluster-robust standard errors; the bootstrap; what robust standard errors can and cannot fix. *Lab*: a review of the calculus used in estimation and optimization; robust and clustered inference and the bootstrap in `R`.

- *Reading*: Effect, §13.3 ("Your Standard Errors Are Probably Wrong"); MHE, Chapter 8.
- *Further reading*: Freedman, David A. 2006. "On the So-Called 'Huber Sandwich Estimator' and 'Robust Standard Errors'." *The American Statistician* 60(4), 299–302; Bertrand, Marianne, Esther Duflo, and Sendhil Mullainathan. 2004. "How Much Should We Trust Differences-in-Differences Estimates?" *Quarterly Journal of Economics* 119(1), 249–275.

Week 3 (Sep 8/10): Maximum Likelihood: Theory

Lecture: The likelihood and log-likelihood; the score function, Hessian, and Fisher information; identification and numerical maximization; the asymptotic properties of the MLE (consistency, asymptotic normality, and efficiency) under regularity conditions; and likelihood-ratio, Wald, and score tests. *Lab*: coding a likelihood and optimizing it numerically in `R`.

- *Reading*: Ward and Ahlquist, Chapters 1–2 and 4.
- *Further reading*: Pawitan, Chapters 2–4.

Week 4 (Sep 15/17): Maximum Likelihood: Applications

Lecture: The generalized linear model framework (linear predictor, link, and likelihood); binary outcomes with logit and probit, now grounded in likelihood theory; interpretation via marginal effects, predicted probabilities, and interaction terms; multinomial and ordered models; count models (Poisson and negative binomial). *Lab*: estimating and interpreting limited-dependent-variable models in `R`. The Thursday session also includes a short introduction to **multilevel models**: the random-intercept model as a GLM with one added variance parameter estimated by maximum likelihood, its two assumptions, the two variances and the intraclass correlation, predictors at both levels, and when clustered OLS or fixed effects are enough.

- *Reading*: ISL, §4.1–4.3; Hanmer, Michael J. and Kerem Ozan Kalkan. 2013. "Behind the Curve: Clarifying the Best Approach to Calculating Predicted Probabilities and Marginal Effects from Limited Dependent Variable Models." *American Journal of Political Science* 57(1), 263–277.
- *Further reading*: Raudenbush, Stephen W. and Anthony S. Bryk. 2002. *Hierarchical Linear Models*, 2nd ed. Sage, Chapters 1–2 and 4; Ward and Ahlquist, Chapter 3; Mize, Trenton D. 2019. "Best Practices for Estimating, Interpreting, and Presenting Nonlinear Interaction Effects." *Sociological Science* 6, 81–117

Week 5 (Sep 22/24): Causal Inference: Potential Outcomes and DAGs

Lecture: The potential outcomes framework; SUTVA, ignorability, and identification; average treatment effects; directed acyclic graphs, back-door and front-door criteria, and collider bias; defining a well-posed estimand. *Lab*: simulating confounding and selection; drawing and reading DAGs in `R` (`dagitty`).

- *Reading*: Effect, Chapters 6 and 8; Lundberg, Ian, Rebecca Johnson, and Brandon M. Stewart. 2021. "What Is Your Estimand? Defining the Target Quantity Connects Statistical Evidence to Theory." *American Sociological Review* 86(3), 532–565; Elwert, Felix and Christopher Winship. 2014. "Endogenous Selection Bias: The Problem of Conditioning on a Collider Variable." *Annual Review of Sociology* 40, 31–53.
- *Further reading*: MHE, Chapter 2; CCI, Chapters 1–3; Winship, Christopher and David J. Harding. 2008. "A Mechanism-Based Approach to the Identification of Age–Period–Cohort Models." *Sociological Methods & Research* 36(3), 362–401.

Week 6 (Sep 29/Oct 1): Matching, Propensity Scores, and Weighting

Lecture: Selection on observables; exact, nearest-neighbor, and coarsened matching; the propensity score; inverse-probability weighting; doubly robust estimation; and what each method assumes when it fails. *Lab*: matching and weighting workflows in `R` (`MatchIt`, `WeightIt`).

- *Reading*: Effect, Chapter 14; CCI, Chapter 5; Stuart, Elizabeth A. 2010. "Matching Methods for Causal Inference: A Review and a Look Forward." *Statistical Science* 25(1), 1–21.
- *Further reading*: CCI, Chapters 4, 7, and 8; King, Gary and Richard Nielsen. 2019. "Why Propensity Scores Should Not Be Used for Matching." *Political Analysis* 27(4), 435–454.

Week 7 (Oct 6/8): Instrumental Variables

Lecture: Endogeneity and the IV estimator; two-stage least squares; the LATE framework, monotonicity, and compliers; weak instruments and sensitivity. *Lab*: 2SLS and diagnostics in `R` (`fixest`/`ivreg`).

- *Reading*: Effect, Chapter 19; MHE, §4.1 and §4.4 (the latter is where LATE and the compliance types live); Angrist, Joshua D., Guido W. Imbens, and Donald B. Rubin. 1996. "Identification of Causal Effects Using Instrumental Variables." *Journal of the American Statistical Association* 91(434), 444–455.
- *Further reading*: CCI, Chapter 9; Lee, David S., Justin McCrary, Marcelo J. Moreira, and Jack Porter. 2022. "Valid t-ratio Inference for IV." *American Economic Review* 112(10), 3260–3290.

Week 8 (Oct 13/15): Fall Break / In-Class Midterm

Tuesday, Oct 13: no class (Fall break, Oct 9–14). *Thursday, Oct 15*: in-class midterm covering Weeks 1–7.

Week 9 (Oct 20/22): Regression Discontinuity Designs

Lecture: Sharp and fuzzy RD; identification at the cutoff; local linear estimation, bandwidth choice, and kernel weighting; validity, sorting, and placebo checks. *Lab*: RD estimation and diagnostics in `R` (`rdrobust`).

- *Reading*: Effect, Chapter 20; Cattaneo, Matias D. and Rocío Titiunik. 2022. "Regression Discontinuity Designs." *Annual Review of Economics* 14, 821–851.
- *Further reading*: MHE, Chapter 6; McCrary, Justin. 2008. "Manipulation of the Running Variable in the Regression Discontinuity Design: A Density Test." *Journal of Econometrics* 142(2), 698–714.

Week 10 (Oct 27/29): Panel Data and Difference-in-Differences

Lecture: Panel data and unobserved heterogeneity; pooled OLS, first-difference, and fixed-effects (within) and random-effects estimators; two-way (unit and time) fixed effects (TWFE); the canonical 2×2 DID and the parallel-trends assumption; DID and event studies as special cases of TWFE; problems with TWFE under staggered adoption and recent robust estimators. *Lab*: panel and DID estimation in `R` (`plm`, `fixest`, `did`).

- *Reading*: Effect, Chapters 16 and 18; Roth, Jonathan, Pedro H.C. Sant'Anna, Alyssa Bilinski, and John Poe. 2023. "What's Trending in Difference-in-Differences? A Synthesis of the Recent Econometrics Literature." *Journal of Econometrics* 235(2), 2218–2244.
- *Further reading*: MHE, Chapter 5; CCI, Chapter 11; Goodman-Bacon, Andrew. 2021. "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics* 225(2), 254–277.

Week 11 (Nov 3/5): Synthetic Control and Its Extensions

Lecture: Comparative case studies; the synthetic control estimator; inference with few treated units; augmented and generalized synthetic control, and synthetic difference-in-differences. *Lab*: synthetic control in `R` (`Synth`, `gsynth`).

- *Reading*: Abadie, Alberto. 2021. "Using Synthetic Controls: Feasibility, Data Requirements, and Methodological Aspects." *Journal of Economic Literature* 59(2), 391–425.
- *Further reading*: Abadie, Alberto, Alexis Diamond, and Jens Hainmueller. 2010. "Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California's Tobacco Control Program." *Journal of the American Statistical Association* 105(490), 493–505; Xu, Yiqing. 2017. "Generalized Synthetic Control Method: Causal Inference with Interactive Fixed Effects Models." *Political Analysis* 25(1), 57–76.

Week 12 (Nov 10/12): Machine Learning: Prediction and Regularization

Lecture: Prediction versus inference; the bias–variance tradeoff; cross-validation; ridge, lasso, and elastic net; trees, random forests, and boosting. *Lab*: regularized regression and tree ensembles in `R` (`glmnet`, `ranger`).

- *Reading*: ISL, §2.1–2.2, 5.1, 6.2, and 8.1–8.2; Molina, Mario and Filiz Garip. 2019. "Machine Learning for Sociology." *Annual Review of Sociology* 45(1), 27–45.
- *Further reading*: Mullainathan, Sendhil and Jann Spiess. 2017. "Machine Learning: An Applied Econometric Approach." *Journal of Economic Perspectives* 31(2), 87–106.

Week 13 (Nov 17/19): Causal Machine Learning

Lecture: Why naive ML is biased for causal parameters; Neyman orthogonality and double/debiased machine learning (DML); partialling-out and cross-fitting; heterogeneous treatment effects and causal forests. *Lab*: DML and heterogeneous effects in `R` (`DoubleML`, `grf`).

- *Reading*: Athey, Susan and Guido W. Imbens. 2017. "The State of Applied Econometrics: Causality and Policy Evaluation." *Journal of Economic Perspectives* 31(2), 3–32; CML, Chapter 4.
- *Further reading*: CML, Chapter 10; Wager, Stefan and Susan Athey. 2018. "Estimation and Inference of Heterogeneous Treatment Effects Using Random Forests." *Journal of the American Statistical Association* 113(523), 1228–1242.

Week 14 (Nov 24): Synthesis and Exam Review

Tuesday, Nov 24: a synthesis of how estimation, causal identification, and machine learning fit together, followed by a review for the final exam: what each design assumes, what it estimates, and how to choose between them. *Thursday, Nov 26*: no class (Thanksgiving recess). The **final exam** takes place on **Thursday, December 3, 9:05–11:35AM**; project-option memos and replication code are due Friday, December 11.